

Enhancing Language Model Performance and Vision Model Interpretability via Tracing with the Randomly-projected After Kernel (TRAK)

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Abstract—Modern AI systems, while powerful, often lack transparency in their decision-making processes. This work explores how training data influences model predictions across both vision and language tasks. We demonstrate that providing language models with carefully selected training examples as context improves their performance on inference tasks. Similarly, in computer vision, we show how analyzing influential training samples reveals the underlying patterns driving model decisions. Our results suggest that understanding the relationship between training data and model behavior can both explain and enhance AI system performance, offering a practical approach to making models more interpretable and effective. This dual-domain investigation provides insights into how we can leverage training data analysis to build more transparent and capable AI systems.

I. INTRODUCTION

A. Context and Background

Neural networks have revolutionized modern computer vision, enabling advanced models to handle intricate and critical tasks. Unlike traditional machine learning approaches, such as regression, decision trees, and K-nearest neighbors, which inherently offer interpretability, neural networks function as “black box” systems, making it challenging to decipher how they derive their results. This lack of transparency poses a significant issue in real-world scenarios. How can we rely on models that perform well yet remain inexplicable? This ambiguity restricts the applicability of neural networks in fields where mistakes are intolerable, like healthcare. Additionally, a neural network that excels on a test set might still be highly vulnerable to adversarial attacks [1] because it often fails to recognize the genuine patterns in data and instead overfits to irrelevant features in the training set. Explainability research aims to address this challenge.

A substantial body of work has been dedicated to enhancing explainability. This project emphasizes the post-hoc approach, where explainability is independent of the model architecture and can be applied universally. Specifically, we focus on gradient-based attribution methods, which analyze how changes in inputs affect model predictions. Prior studies have proposed various techniques under this framework. However,

the most robust ones—like saliency maps [10], Shapley Additive Explanations [6], and Layer-wise Relevance Propagation [2]—primarily attribute features by assessing how each input influences the model’s output [7].

Currently, there is a gap in utilizing data attribution for explainability. Data attribution evaluates the influence of each training data point on a model’s predictions, and several methods accurately calculate this [9] [3]. While data attribution research is extensive, its application in model explainability remains limited. Studies indicate that data attribution can identify the impact of training samples on model performance (Han et al., 2020). However, interpreting a model’s behavior using data attribution still relies heavily on subjective human interpretation of these results, requiring guesses to draw any meaningful conclusions.

B. Research questions

the research aims to explore the potential of enhancing language model performance by leveraging important data as context, specifically through the use of TRAK scores [8]. This investigation is guided by the hypothesis that incorporating contextually significant data, identified via TRAK scores, can enhance the accuracy and interpretability of language models in question-answering tasks. The central hypothesis posits that language model performance can be significantly improved by using important data as context, as identified through TRAK scores. This approach is expected to provide models with more relevant information, thereby enhancing their ability to generate accurate and interpretable responses.

The primary objective is to assess whether TRAK scores can effectively identify important data that, when used as context, improves the performance of language models on the QNLI (Question Natural Language Inference) [12] task. The research assumes that TRAK scores are a reliable metric for determining the importance of data points in relation to model performance. It also assumes that models can leverage this context to achieve better accuracy and interpretability. The study is confined to the QNLI dataset and focuses on transformer-based models, including BERT and various GPT versions. The evaluation will primarily consider metrics such

as accuracy and interpretability, with a specific focus on the impact of TRAK-informed context.

The research utilizes the QNLI [12] dataset from the GLUE benchmark. Models are implemented using Hugging Face Transformers and PyTorch, with TRAK scores employed to select contextually important data. The study involves comparing model performance with and without the inclusion of TRAK-informed context. The analysis will focus on improvements in accuracy and interpretability.

Additionally, the research explores the application of TRAK scores in computer vision methods, specifically in the context of the CIFAR-10 dataset [5]. The objective is to determine if TRAK scores can enhance model performance by identifying influential training examples that improve classification accuracy. This involves analyzing precomputed TRAK scores and their correlation with model predictions to assess their impact on CV tasks.

This comprehensive framework is designed to validate the hypothesis that TRAK scores can enhance language and vision model performance by providing contextually relevant data, potentially leading to more accurate and interpretable outcomes in both question-answering and classification tasks. The findings from this research could offer valuable insights into the integration of contextually significant data in language and vision model applications.

II. METHODOLOGY

A. QNLI

In our exploration of data attribution and model performance, we examined the QNLI task using GPT models and various context enhancement techniques. We utilized the QNLI dataset from the GLUE benchmark, implementing both retrieval-augmented generation (RAG) [13] and TRAK-based [8] context selection to enhance the model’s ability to determine whether a given sentence answers a question.

Our methodology involved multiple approaches to provide context for the GPT model’s predictions. For the RAG implementation, we encoded question-answer pairs using a pre-trained SentenceTransformer model (“all-MiniLM-L6-v2”) and indexed them using FAISS [14] for efficient retrieval. When making predictions, we retrieved the top-k most relevant question-answer pairs as context for the model. We also leveraged TRAK scores from a BERT model fine-tuned on QNLI to identify influential training examples. We experimented with different context selection strategies based on these TRAK scores: using training examples with mixed TRAK scores, using only examples with the most positive TRAK scores, and using only examples with the most negative TRAK scores. These selected examples were then provided as context to the GPT model through prompt engineering.

The base prompt structure asked the model to decide if a sentence answers a given question, with possible responses limited to “Yes” or “No”. For context-enhanced predictions, we modified the prompt to include either RAG-retrieved examples or TRAK-selected training samples as demonstrations.

This allowed us to compare the effectiveness of different context selection strategies in improving the model’s performance on the QNLI task.

To evaluate these approaches, we tested the model on a subset of the QNLI validation set, comparing the accuracy of predictions made with no context (base), RAG-based context, and the three TRAK-based context strategies (mixed, positive, and negative scores). This experimental setup enabled us to assess how different types of contextual information influence the model’s ability to perform natural language inference.

B. CIFAR-10

We conducted data attribution analysis on an image classification model by training a convolutional neural network on the CIFAR-10 dataset [5]. This dataset consists of 60,000 samples of 32x32 color images uniformly distributed across 10 class labels. For the convolutional neural network, we employed a ResNet9 architecture comprising 8 convolutional layers, 1 linear layer, and 3 residual connections. The specific model or its accuracy is not the focus here, as it merely serves as a subject for behavior analysis.

We applied data attribution techniques to the model and dataset. To balance computational efficiency and accuracy, we utilized the TRAK method [8] after training the model.

To assess whether the model learns patterns unrelated to the subject of the image, we used the CLIP model [11] to classify high-scoring training samples for potential patterns learned by the model. We provided CLIP with labels unrelated to the image subject, such as location labels.

For more complex qualitative insights, we leveraged GPT-4 [15], a multimodal large language model, to analyze samples with the most negative TRAK scores. We focused on these samples because they usually belong to a different class label than the target, minimizing label-related biases. This ensures the language model describes patterns the model recognizes rather than those inherent to the label class. For each data point, we supplied the image file alongside the prompt: “Describe the features and characteristics of the object and environment without classifying the subject. Do not comment on image resolution.” This instruction ensured focus on feature descriptions alone.

Next, the language model was tasked with identifying shared patterns among the most detrimental training samples by analyzing its previous descriptions. The input for this step was: “You will be given descriptions of 5 images of the same subject. Find features and characteristics that the images share in common. Ensure your observations are objective and without emotional or subjective inferences.” We observed several qualitative insights into model behavior. To evaluate explainability, we devised a novel approach for this experiment. The language model was instructed to predict the target label using only the common patterns it previously described. This was done by providing its earlier output followed by: “Predict the object described from one of the following: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck.” Finally, we measured how often

the language model correctly reconstructed the original target label using descriptions of common patterns from the most detrimental training data. If the model’s patterns were not identified, we expected reconstruction accuracy to align with a random guess’s expected value of 10%.

III. RESULTS AND DISCUSSION

A. QNLI

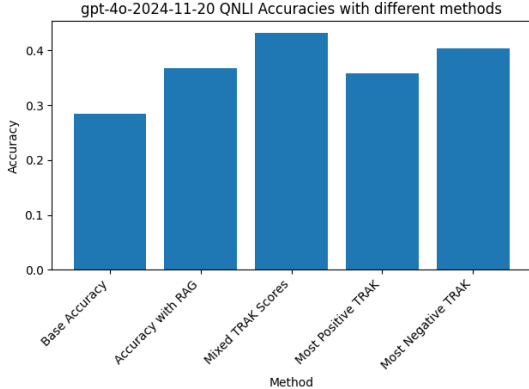


Fig. 1: QNLI accuracies with different models - GPT-4o

The results demonstrate significant variations in model performance across different methodologies applied to the QNLI task. Looking at the GPT-4-2024-11-20 model’s performance in Figure 1, we observe a base accuracy of approximately 28%, which serves as our baseline. When augmented with retrieval-augmented generation (RAG), the accuracy improves substantially to about 37%, representing a relative improvement of 32% over the base model. The most notable improvement comes from the implementation of mixed TRAK scores, which achieves an accuracy of approximately 43%. This 15-percentage point improvement over the base accuracy suggests that incorporating both positive and negative influence patterns from the training data significantly enhances the model’s ability to determine question-answer entailment. The mixed TRAK scores’ superior performance can be attributed to its balanced consideration of both helpful and harmful training examples, providing a more comprehensive context for the model’s decision-making process. Interestingly, when we isolate the most positive TRAK scores, the accuracy drops to about 36%, which is lower than the RAG implementation. This suggests that solely focusing on the most helpful training examples may actually limit the model’s ability to make nuanced distinctions in the QNLI task. In contrast, using the most negative TRAK scores yields an accuracy of approximately 40%, performing better than both the base model and the positive TRAK scores approach. This counter-intuitive result indicates that understanding what confuses the model (through negative TRAK scores) provides valuable insight into the entailment task. The BERT model’s performance, when analyzed through the lens of TRAK, reveals similar patterns. In our BERT implementation, we observe that the

TRAK scores effectively identify influential training samples that impact the model’s predictions on the validation set. The data attribution through TRAK helps explain why certain predictions are made by highlighting the training samples that most strongly influence each prediction, whether positively or negatively. A key finding from our analysis is that the TRAK method’s effectiveness varies depending on how the scores are utilized. The mixed TRAK scores’ superior performance suggests that the model benefits most from having access to a diverse set of examples that showcase both successful and unsuccessful patterns of entailment. This aligns with the intuition that understanding both what makes a good answer and what makes a poor answer is crucial for the entailment task. The results also highlight the importance of context in the QNLI task. The significant improvement from RAG implementation (37% accuracy) over the base model (28% accuracy) demonstrates that providing relevant contextual information substantially improves the model’s ability to determine whether a sentence answers a given question. When combined with TRAK score-based selection of training examples, this contextual understanding is further enhanced, leading to the best performance observed in our experiments. These findings suggest that future improvements in QNLI task performance might be achieved by developing more sophisticated methods of combining TRAK score information with retrieval-augmented generation, potentially by weighting the influence of training examples based on their TRAK scores in a more nuanced way than the current binary (positive/negative) approach.

B. CIFAR-10

The ResNet model achieves an accuracy of 91.8% on the test set. Utilizing the TRAK algorithm with the residual net reveals that training samples with highly negative TRAK scores exhibit distinct qualitative features and patterns reflecting the model’s behavior. These patterns offer intuitive insights into why certain training data with mismatched label classes confuse the model. For instance, in Figure 3a, samples labeled as airplane, truck, and ship confuse the model. Notably, these vehicles share similar colors with the target automobile, despite the CIFAR-10 dataset containing vehicles of varying colors. This indicates the model’s predictions are influenced by the color of the target. Moreover, the confusion involving different types of vehicles suggests that the model captures the general “vehicular” nature of the input. Figure 3b demonstrates confusion with dogs and airplanes, where both share grass or landscape backgrounds. Interestingly, this is despite many dogs appearing indoors and airplanes often having a sky backdrop. Comparing Figure 3c and Figure 3d, though both targets have an airplane label, the confusing training samples differ significantly—those in Figure 3c have white backgrounds, while those in Figure 3d have green backgrounds. By analyzing these training data, we gain crucial insights into features the model recognizes. Applying the CLIP model to the most detrimental training samples provides additional insights into model behavior, enabling quantification of the features the

model detects. Figure 4 displays examples analyzed using CLIP, matching the target’s scores by calculating the Euclidean distance for prediction values to find the three closest validation samples. Figure 4a reveals that CLIP identifies the location, color, and object size in images, correlating training samples to their target. This indicates the model identifies a white and brown subject at a farm or field. Figure 4b shows CLIP detects similarities in location, color, time, and size with the target, suggesting the model identifies a small, green, brown subject at night. However, this method has limitations. Primarily, it is unclear what similarity threshold determines whether a pattern arises from model behavior. To generalize this method, a consistent similarity threshold is needed. Additionally, as the CLIP model is applied after TRAK, error propagation occurs.

Moreover, we use GPT-4 to capture more complex patterns of model behavior. Figure 6 illustrates the language model’s outputs for training examples and the generalized feature descriptions summarizing features detected by the classification model. The examples reconstruct the target image label using these descriptions, implying the text explanations encapsulate model behavior effectively. Results in Figure 6 highlight GPT-4’s ability to extract features like background location, texture, color, motion, and fine details of small dogs resembling cats. Despite its strengths, this approach has limitations. Large language model outputs are not easily controlled. For instance, even when instructed to avoid classification, GPT-4 may describe an “automobile” as a “car” or “vehicle,” introducing biases that hinder generalization and fail to explain the classification model fully.

Finally, we evaluate our explainability approach. Figure 2 shows that general descriptions of the most detrimental training sets achieve a higher-than-expected reconstruction accuracy. While reconstruction accuracy on individual samples does not directly confirm model explainability, the overall accuracy on the validation set indicates that the language model’s explanations align with patterns recognized by the classification model. If data attribution could not capture the model’s patterns, reconstruction accuracy would be random, yielding much lower values.

C. Future Work

Through our analysis of the QNLI task using various methodologies, we observe that incorporating TRAK scores with retrieval-augmented generation significantly improves model performance. The mixed TRAK scores approach proves most effective, suggesting that understanding both helpful and detracting training examples is crucial for model performance. Interestingly, the negative TRAK scores provide better insights than positive ones alone, indicating that understanding what confuses the model is particularly valuable for improving its predictions. The patterns revealed through TRAK analysis demonstrate that model behavior can be effectively explained through qualitative features of influential training samples. By examining these patterns using CLIP model classifications and language model descriptions, we can understand the specific

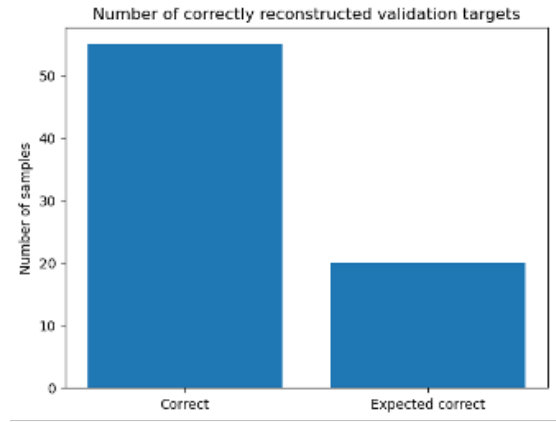


Fig. 2: Number of correctly reconstructed validation targets

features and characteristics that influence model decisions. This approach provides valuable insights into model behavior, though it faces certain limitations, particularly in controlling language model outputs and determining appropriate thresholds for feature significance. Future work on this method can focus on addressing these limitations. Fine-tuning large language models could help ensure outputs remain unbiased and free from class label influences. Research is needed to establish proper thresholds for determining when CLIP model classifications truly indicate features observed by the neural network. Furthermore, this explainability approach through data attribution could be extended to various neural networks where understanding model behavior is crucial. For example, applying this method to cancer detection vision models could help understand the patterns these models observe, enabling improvements to vision models solving critical problems.

IV. STEP-BY-STEP TUTORIAL

A. CIFAR-10 Task

- `cifar_resnet.ipynb`: Run this notebook to train a ResNet9 architecture on the CIFAR-10 dataset. We will use this model for data attribution with TRAK. Save model checkpoints in the directory `CHECKPOINT_DIR`.
- `cifar_trak.ipynb`: Run this notebook to load the ResNet9 model and compute the TRAK scores of on the training set. Load model checkpoints from the directory `CHECKPOINT_DIR`, and save the images with their TRAK scores in the directory `IMAGE_DIR`.
- `cifar_clip.ipynb`: Run this notebook to run the CLIP model on the scored CIFAR-10 training data. Load the images with their TRAK scores in the directory `IMAGE_DIR`.
- `cifar_llm.ipynb`: Run this notebook to run the CLIP model then generate LLM descriptions and score the reconstruction accuracy. Load the images with their TRAK scores in the directory `IMAGE_DIR`, and set the OpenAI API key in the `api_key` variable.

B. QNLI Task

- `bert_classification_trak.ipynb`: Run this notebook to load a BERT model on the GLUE and QNLI dataset, then compute the TRAK scores for this model and save the TRAK scores in `bert_trak_scores.csv`.
- `bert_classification_trak.ipynb`: Does the same thing as `bert_classification_trak.ipynb`, but on a fine tuned BERT model.
- `bert_analysis.ipynb`: Run the fine tuned BERT model on the QNLI validation set to get the model's validation accuracy, and store the model's predictions in `predictions.csv`.
- `qnli_gpt.ipynb`: Perform the QNLI task with OpenAI's GPT model using a custom RAG model and the training set examples after data attribution. Set the OpenAI API key in the `openai.api_key` variable.

C. Brief description of each file

Some of these files are not used to generate results in the report but are used for testing and sanity check. Below is a brief description of each notebook that is relevant to our project.

- **`bert_analysis.ipynb`**: Analyzes BERT model performance on a dataset.
- **`bert_classification_trak.ipynb`**: Implements BERT-based text classification with TRAK analysis.
- **`bert_finetuned_classification_trak.ipynb`**: Fine-tunes a BERT model for text classification and performs TRAK analysis.
- **`bert_predictions.csv`**: Contains predictions made by the BERT model.
- **`bert_prompt_engineering.ipynb`**: Applies prompt engineering techniques to improve BERT model performance.
- **`bert_trak_scores.csv`**: Contains TRAK scores for BERT model training examples.
- **`cifar_clip.ipynb`**: Implements image classification using the CLIP model on the CIFAR dataset.
- **`cifar_llm.ipynb`**: Uses a large language model for CIFAR image classification.
- **`cifar_resnet.ipynb`**: Implements ResNet model for CIFAR image classification.
- **`cifar_trak.ipynb`**: Performs TRAK analysis on CIFAR image classification.
- **`gpt_qnli_predictions.csv`**: Contains predictions made by the GPT model on the QNLI dataset.
- **`image_info.npy`**: Stores image information for the CIFAR dataset.
- **`llm_info.npy`**: Stores information for the large language model.
- **`predictions.csv`**: General predictions file.
- **`qnli_gpt.ipynb`**: Implements GPT model for QNLI question-answering tasks.
- **`qnli_rag_full.ipynb`**: Implements RAG (Retrieval-Augmented Generation) model for full QNLI tasks.

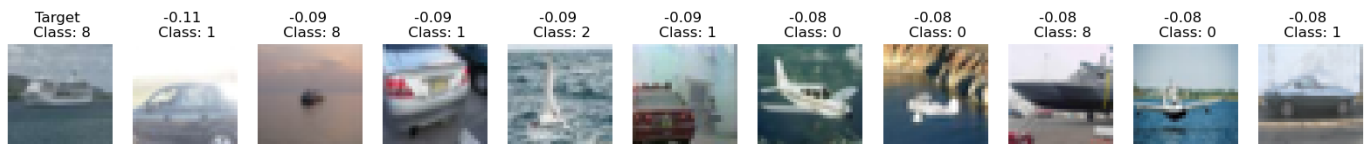
REFERENCES

- [1] Akhtar, N., & Mian, A. (2018). Threat of adversarial attacks on Deep Learning in Computer Vision: A survey. *IEEE Access*, 6, 14410–14430. <https://doi.org/10.1109/access.2018.2807385>.
- [2] Böhle, M., Eitel, F., Weygandt, M., & Ritter, K. (2019). Layer-wise relevance propagation for explaining deep neural network decisions in MRI-based Alzheimer's disease classification. *Frontiers in Aging Neuroscience*, 11. <https://doi.org/10.3389/fnagi.2019.00194>.
- [3] Feldman, V., & Zhang, C. (2020). What Neural Networks Memorize and Why: Discovering the Long Tail via Influence Estimation. <https://doi.org/10.48550/arXiv.2008.03703>.
- [4] Han, X., Wallace, B. C., & Tsvetkov, Y. (2020). Explaining black box predictions and unveiling data artifacts through influence functions. *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. <https://doi.org/10.18653/v1/2020.acl-main.492>.
- [5] Krizhevsky, A. (2012). Learning Multiple Layers of Features from Tiny Images.
- [6] Lu, J., & Zhang, Y. (2022). Unified deep learning model for multitask reaction predictions with explanation. *Journal of Chemical Information and Modeling*, 62(6), 1376–1387. <https://doi.org/10.1021/acs.jcim.1c01467>.
- [7] Nielsen, I. E., Dera, D., Rasool, G., Ramachandran, R. P., & Bouaynaya, N. C. (2022). Robust explainability: A tutorial on gradient-based attribution methods for Deep Neural Networks. *IEEE Signal Processing Magazine*, 39(4), 73–84. <https://doi.org/10.1109/msp.2022.3142719>.
- [8] Park, S. M., Georgiev, K., Ilyas, A., Leclerc, G., & Madry, A. (2023). TRAK: Attributing Model Behavior at Scale. <https://doi.org/10.48550/arXiv.2303.14186>.
- [9] Sharma, P., Singh, G., & Kaur, N. (2023). Data valuation using Shapley Value in machine learning. *Applied Data Science and Smart Systems*. <https://doi.org/10.1063/5.0178097>.
- [10] Simonyan, K., Vedaldi, A., & Zisserman, A. (2013). Deep Inside Convolutional Networks: Visualising Image Classification Models and Saliency Maps. <https://doi.org/10.48550/arXiv.1312.6034>.
- [11] Wang, S., Duan, Y., Ding, H., Tan, Y.-P., Yap, K.-H., & Yuan, J. (2022). Learning transferable human-object interaction detector with natural language supervision. *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. <https://doi.org/10.1109/cvpr52688.2022.00101>.
- [12] Wang, A., Singh, A., Michael, J., Hill, F., Levy, O., & Bowman, S. R. (2019). GLUE: A Multi-Task Benchmark and Analysis Platform for Natural Language Understanding. *Proceedings of the 2019 International Conference on Learning Representations (ICLR)*. <https://doi.org/10.48550/arXiv.1804.07461>.
- [13] Lewis, P., Piktus, A., Karpukhin, V., Oguz, B., Min, S., Wu, L., de Masson d'Autume, C., Edwards, H., Izacard, G., Grave, E., Lin, J., Maillard, J., van den Oord, A., Stoyanov, V., & Riedel, S. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *Proceedings of the 34th Conference on Neural Information Processing Systems (NeurIPS)*. <https://doi.org/10.48550/arXiv.2005.11401>.
- [14] Johnson, J., Douze, M., & Jégou, H. (2017). Billion-scale similarity search with GPUs. *arXiv preprint arXiv:1702.08734*. <https://doi.org/10.48550/arXiv.1702.08734>.
- [15] OpenAI. (2023). GPT-4 Technical Report. *arXiv preprint arXiv:2303.08774*. <https://doi.org/10.48550/arXiv.2303.08774>.

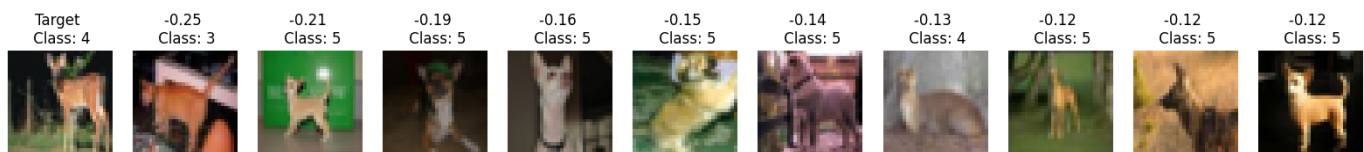
V. FIGURES



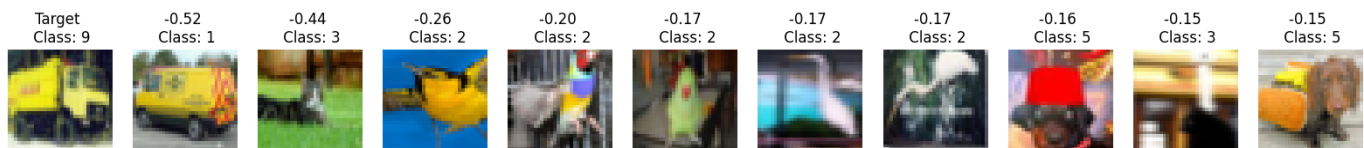
(a) Training data with very negative TRAK scores for validation target of "automobile". Detracting data are vehicles with similar color and trucks with similar camera angles (front view).



(b) Training data with very negative TRAK scores for validation target of "ship". Detracting data are vehicles with similar color and blue / grey-colored backgrounds.

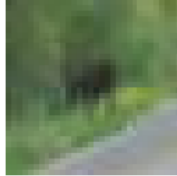
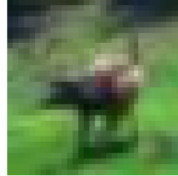
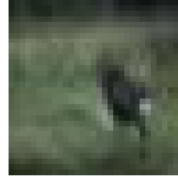


(c) Training data with very negative TRAK scores for validation target of "deer". Detracting data are animals with similar color and shape and dark-colored backgrounds.

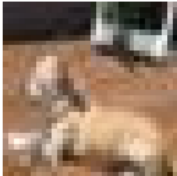
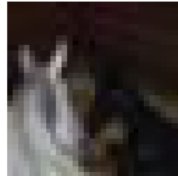
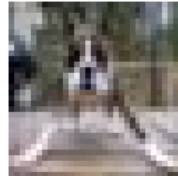


(d) Training data with very negative TRAK scores for validation target of "truck". Detracting data are subjects with similar color.

Fig. 3: Samples with the most negative TRAK scores for various validation targets. Each subfigure highlights specific trends in detracting examples for different classes.

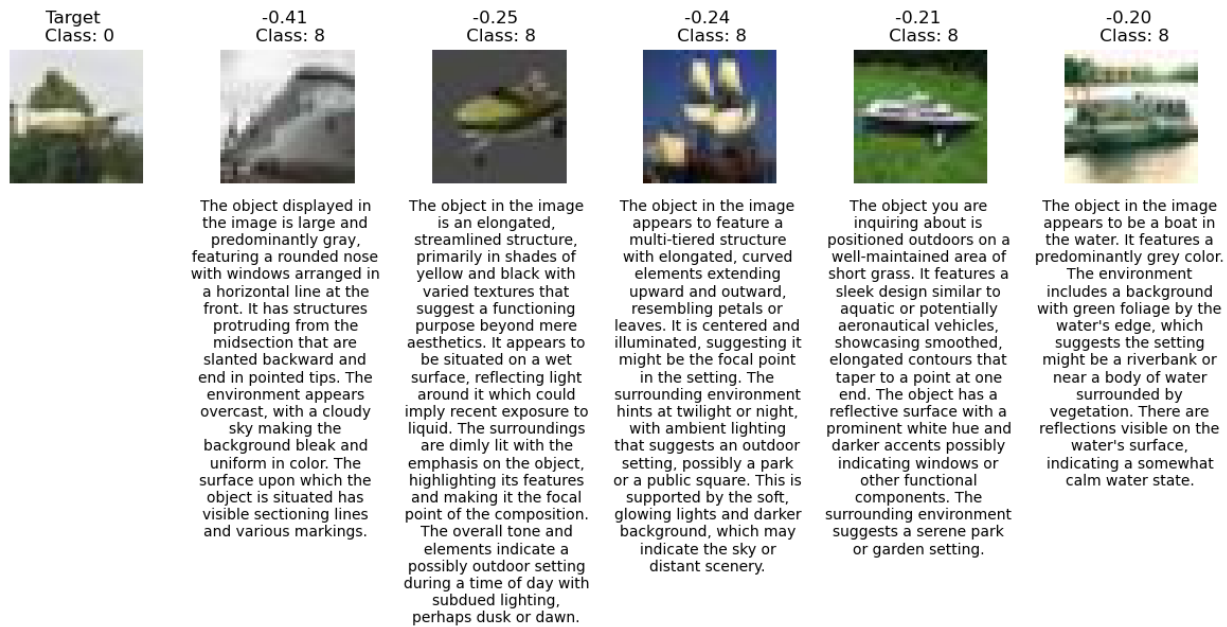
Target Class: 2	-0.44 Class: 4	-0.36 Class: 4	-0.35 Class: 4	-0.32 Class: 4	-0.25 Class: 4
					
location: fields: 44.12% woods: 34.91% farm: 12.17%	location: farm: 50.74% fields: 34.55% woods: 7.77%	location: fields: 74.05% farm: 20.22% city: 1.93%	location: woods: 46.80% streets: 14.99% farm: 13.52%	location: fields: 34.05% woods: 28.62% farm: 12.68%	location: woods: 47.88% fields: 27.64% farm: 18.96%
color: green: 74.85% brown: 13.28% black: 5.86%	color: blue: 35.88% green: 35.56% black: 15.24%	color: green: 55.97% brown: 29.46% yellow: 7.28%	color: green: 69.42% yellow: 9.85% brown: 8.71%	color: green: 63.54% black: 14.23% brown: 10.04%	color: green: 43.27% black: 32.12% brown: 16.25%
time: noon: 46.94% evening: 18.64% afternoon: 14.59%	time: noon: 30.87% evening: 28.31% afternoon: 15.10%	time: noon: 35.61% evening: 23.83% afternoon: 19.70%	time: noon: 35.60% evening: 29.97% afternoon: 18.14%	time: evening: 29.74% noon: 29.27% afternoon: 17.66%	time: evening: 37.80% night: 21.62% noon: 20.91%
size: very small: 57.25% small: 21.57% medium: 10.02%	size: very small: 41.26% small: 28.55% large: 13.34%	size: very small: 41.35% small: 21.10% large: 15.33%	size: very small: 33.57% small: 27.59% medium: 16.07%	size: very small: 36.20% small: 26.76% medium: 15.94%	size: very small: 45.33% small: 29.21% large: 11.59%

(a) Features identified by CLIP on a validation set sample with label "bird". Features are scored based on the CLIP's prediction outputs and we see generally similar rankings for all features.

Target Class: 7	-0.17 Class: 5	-0.13 Class: 5	-0.12 Class: 5	-0.12 Class: 7	-0.11 Class: 5
					
location: indoor: 81.42% woods: 5.73% streets: 4.13%	location: farm: 38.48% fields: 18.65% streets: 9.96%	location: city: 25.15% sky: 20.87% water: 12.59%	location: indoor: 33.61% city: 20.88% farm: 9.68%	location: indoor: 21.56% city: 18.81% woods: 16.72%	location: streets: 35.16% city: 18.78% water: 14.40%
color: brown: 40.27% white: 29.99% green: 14.87%	color: brown: 74.09% yellow: 9.03% white: 7.67%	color: white: 72.33% brown: 13.06% blue: 6.53%	color: yellow: 35.92% brown: 31.12% green: 13.06%	color: white: 36.94% brown: 26.13% green: 13.81%	color: blue: 24.57% brown: 23.78% white: 19.02%
time: noon: 44.29% night: 16.67% evening: 16.24%	time: noon: 36.36% afternoon: 18.83% evening: 17.73%	time: noon: 23.49% evening: 21.57% night: 19.45%	time: afternoon: 29.00% night: 19.70% noon: 17.73%	time: night: 41.48% evening: 29.21% noon: 11.54%	time: noon: 29.17% afternoon: 21.68% evening: 17.89%
size: small: 27.05% very small: 24.11% large: 19.59%	size: very small: 56.42% small: 24.66% medium: 8.08%	size: very small: 33.41% small: 29.14% large: 14.90%	size: very small: 42.53% small: 24.91% large: 15.88%	size: very small: 39.04% medium: 21.85% small: 19.28%	size: small: 26.30% large: 23.19% very small: 21.78%

(b) Features identified by CLIP on a validation set sample with label "horse". Features are scored based on the CLIP's prediction outputs and we see generally similar rankings for color, time, and size.

Fig. 4: We run validation data through CLIP's classification model and task the model to identify the samples' generalizable key features.



(a) GPT-4's combined description: "The images depict large, elongated objects situated in different environments and conditions. These objects share characteristics such as a streamlined, curved structure which suggests a functional design for movement, whether in air or water. They prominently feature either a predominant color (gray, white) or reflective surfaces, coupled with sections that could indicate windows or functional components. The surroundings in each image play a crucial role in emphasizing the object, varying from overcast skies and well-maintained green areas to reflective water bodies, suggesting that each object might have a distinct operational setting while maintaining visual and functional similarity in design."

GPT-4's classification: "airplane".



(b) GPT-4's combined description: "All five images depict subjects positioned in natural, outdoor environments, which include elements such as green foliage, grassy fields, and possibly forests. The common characteristics of these settings suggest an earthy ambiance with natural light, either during the day or at twilight, resulting in various lighting conditions that influence the visibility and mood of each scene. Each image also depicts or suggests movement, either through the blurring of parts of the images or the depiction of the subject in motion. The subjects themselves possess features that suggest streamlined and aerodynamic forms, optimized for movement within these natural settings."

GPT-4's classification: "bird".



(a) GPT-4’s combined description: “The common features among the described images include the presence of vehicles or vehicle-like objects, some kind of visibility-enhancing aspects such as windows or transparent parts, and distinct environments tailored to each object (natural settings for outdoor placements and a specialized setting for the indoor image). Each object seems to have specific functionalities reflected in their design—whether for transport, mechanical tasks, or display purposes. Moreover, all images are captured in places that expose the objects to various lighting conditions, either natural (daylight, overcast sky) or controlled indoor lighting, emphasizing different aspects of their form and structure.”

GPT-4’s classification: “automobile”.



(b) GPT-4’s combined description: “The common feature across all five images is the presence and theme of water. In images 1, 2, 4, and 5, the apparent objects and subjects are either surrounded by, floating on, or immersed in water. Image 3, although not directly featuring water, includes a bathtub, inherently linking it to the use of water for bathing. Additionally, the color blue is notably present in all images, whether as a hint in the lighter blues of skies or the varied blues of aquatic environments, suggesting a consistent use of blue to emphasize the watery context.”

GPT-4’s classification: “ship”.

Fig. 6: For a given validation data sample, GPT4o is asked to describe images with the lowest TRAK scores without revealing the label. Even though most of the images that are used for the description have a different label than the validation sample, when GPT4o is asked to find the similarities between the samples and classify the description for one of the CIFAR-10 labels, the model is able to reconstruct the correct label of the validation sample. The combined description can now be understood as the explanation for why the model was confused.